

VibeSensor Diagnostic Report

Run date
2026-03-31 13:16:52 UTC

Sensors
4

Car
Unknown

Speed Band
80-90 km/h

Duration
01:17.9

Suspected source

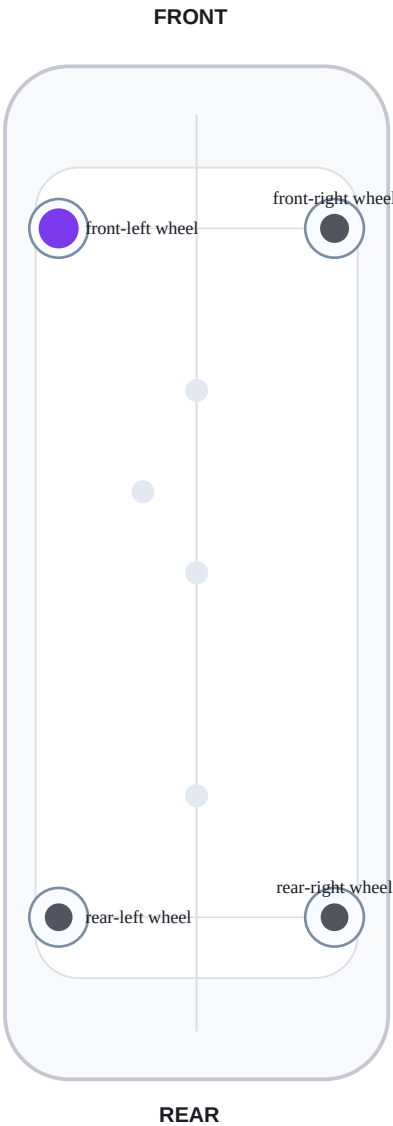
Wheel / Tire

Inspect first: **Front-Left**

Action status

Inspect with caution

Why this corner wins



Front-Left outranked the next location by 3.25x on matched-window linear intensity evidence.

Dominant corner

Front-Left

Location confidence

Limited by run context

Coverage

4 of 4 wheel positions seen throughout run.

Alternative source

Driveline

What to do next

- 1 Check Front-Left for tire damage, flat spots, belt shift, uneven wear, or pressure
- 2 Check Front-Left for imbalance or radial/lateral runout.
- 3 Check driveline mounts and couplings near Front-Left for looseness or cracked rubber.

Evidence and Run Context

Evidence chain

Sensor matches count matches across all connected sensors.

Source	Signal	Measurement refs	Sensor matches	Speed window	Location
Wheel / Tire	2x wheel, 1x wheel	M01, M02, M03, M04	938	80-90 km/h	Front-Left

Supporting measurements

Ranked peak rows: Wheel / Tire · 2x wheel, 1x wheel · Front-Left · 80-90 km/h

Measurement	Peak dB	Strength dB
M01	43.9 dB	43.8 dB
M02	34.3 dB	33.9 dB
M03	29.4 dB	28.8 dB
M04	25.0 dB	24.3 dB

Supporting context

Speed-band summary

80-90 km/h

Phase summary

Cruise

Suitability detail

Speed variation

Speed range stayed in a usable diagnostic band for steady-state diagnosis and order tracking.

Sensor coverage

Multiple sensor locations observed.

Reference completeness

Required order references are present.

Saturation and outliers

No obvious saturation detected.

Frame integrity

No dropped frames or queue overflows detected.

Traceability

Run context

Run ID

4a07a7465ee84bd48937c97b5489d784

Run date

2026-03-31 13:16:52 UTC

Sensor Model

ADXL345

Firmware Version

None

Analysis rows

978

Raw Sample Rate (Hz)

800

tire size

285/30R21

Export package

4a07a7465ee84bd48937c97b5489d784_export.zip

Report version

v0.0.0-dev

Inspection Path

Primary vs alternative source

Primary source

Wheel / Tire

Inspect first

Front-Left

Alternative source

Driveline

If the primary path is clean

Move to the Driveline path next and inspect Front-Left.

Action matrix

Action	Why	Confirm	Rule out
Check Front-Left for tire damage, flat spots, belt shift, uneven wear, or pressure mismatch.	Tire structural issues often create strong 1x/2x wheel-order vibration.	Visible/measurable tire defect aligns with complaint speed band.	No tire condition anomaly is found on inspected wheels.
Check Front-Left for imbalance or radial/lateral runout.	Wheel-order signatures are most likely caused by imbalance, runout, or tire deformation.	A clear imbalance or runout is found and corrected, with vibration complaint reduced.	Balance and runout are within spec on all checked wheels/tires and complaint remains unchanged.
Check driveline mounts and couplings near Front-Left for looseness or cracked rubber.	Loose or degraded mounts can amplify normal order content into cabin vibration.	Loose mount/fastener or cracked rubber support is found.	All inspected mounts and fasteners are within condition/torque spec.